

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

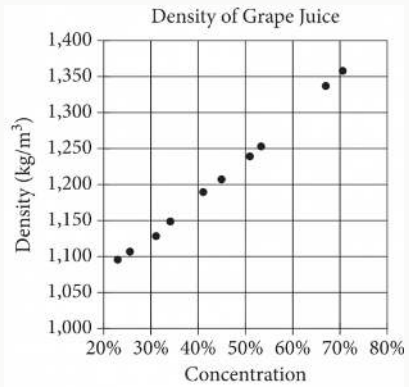
Problem-Solving and Data Analysis

10

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS



The densities of different concentrations of grape juice are shown in the scatterplot above. According to the trend shown by the data, which of the following is closest to the predicted density, in kilograms per cubic meter (kg/m^3), for grape juice with a concentration of 60%?

- A. 1,200
- B. 1,250
- C. 1,300
- D. 1,350

PROBLEM-SOLVING AND DATA ANALYSIS

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

3 / 12

During a sale, the original prices of all the items in a clothing store have been reduced by 20%. What is the sale price of a jacket with an original price of \$50 ?

- A. \$12
- B. \$30
- C. \$36
- D. \$40

Each of 157 gemstones can be classified as one of three classifications, as shown in the frequency table.

Classification	Frequency
color X	119
color Y	3
color Z	35

If one of the gemstones is selected at random, what is the probability of selecting a gemstone of color Y?

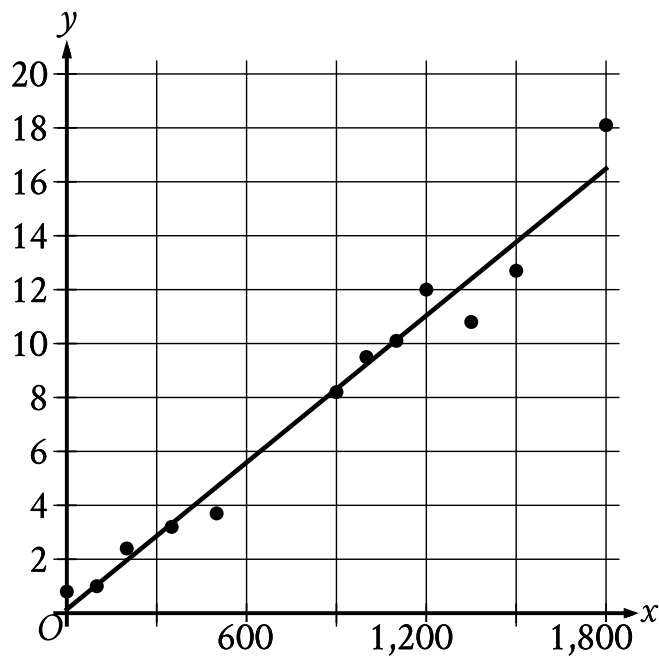
- A. $\frac{3}{157}$
- B. $\frac{35}{157}$
- C. $\frac{119}{157}$
- D. $\frac{154}{157}$

The table below shows the high and low temperatures in Houston, Texas, during a five-day period.

Temperatures in Houston, Texas (degrees Fahrenheit)					
	Monday	Tuesday	Wednesday	Thursday	Friday
High temperature	73	56	62	75	81
Low temperature	49	37	41	54	63

What was the mean low temperature, in degrees Fahrenheit, during the five-day period?

- A. 48.8
- B. 49
- C. 59
- D. 59.1



- The scatterplot has 12 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - The line of best fit passes through the following approximate coordinates:
 - (600 comma 5.6)
 - (1,200 comma 11.0)
 - (1,500 comma 13.8)
 - (1,800 comma 16.5)

Twelve data points are shown in the scatterplot. A line of best fit for the data is also shown. At $x = 1,200$, which of the following is closest to the y -value predicted by the line of best fit?

A. 16

B. 14

C. 11

D. 6

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

An object travels at a constant speed of 6 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 24 centimeters?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

What is 23% of 100?

- A. 23
- B. 46
- C. 77
- D. 123

Colors of Marbles in a
Bag

Color	Number
Red	8
Blue	10
Green	22
Total	40

The table shows the number of different colors of marbles in a bag. If a marble is chosen at random from the bag, what is the probability that the marble will be blue?

- A. $\frac{30}{40}$
- B. $\frac{22}{40}$
- C. $\frac{18}{40}$
- D. $\frac{10}{40}$

4, 4, 4, 4, 8, 8, 8, 13, 13

Which frequency table correctly represents the data listed?

A.

Number	Frequency
4	4
8	3
13	2

B.

Number	Frequency
4	4
3	8
2	13

C.

Number	Frequency
4	16
8	24
13	26

D.

Number	Frequency
16	4
24	8
26	13